

JULIEN SLUK

Backend-Focused Software Developer | Automation Engineer | Lead Trainer

Singapore | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

Professional Summary

Information Technology graduate from Ngee Ann Polytechnic with experience in backend development, automation engineering, monitoring systems, workflow automation, and technical training. Completed a one-year automation internship at OCBC Bank, developing Python and Selenium solutions for operational monitoring and process automation. Built portfolio projects including PulseWatch, a FastAPI monitoring platform with response-time history charts, and a Food Donation Platform with matching logic, machine learning demand prediction, and Leaflet map visualization. Career goal: backend-focused software developer specializing in automation systems, monitoring platforms, and operational tooling.

Technical Skills

Programming: Python, JavaScript, C#, SQL, HTML, CSS

Backend & APIs: FastAPI, REST APIs, API design, CRUD operations, JSON request/response handling

Automation: Selenium, Excel VBA, Google Apps Script, UiPath, Telegram Bots, workflow automation

Data & ML: SQLite, MongoDB, SQL databases, scikit-learn, OpenCV, clustering, prediction models

Tools & Platforms: Git, Docker, Power BI, VS Code, Visual Studio, AWS, Azure, Windows, Linux, SAP HR systems exposure

Professional Experience

Zenitane — Lead Trainer for Computer Science | 2026–Present

- Deliver computer science and technology training programs covering programming, computational thinking, web development, automation, AI tools, and hands-on project building.
- Develop lesson materials, project-based learning activities, and technical workshops for learners across different levels.
- Mentor students in software development projects, debugging, problem-solving, and practical technology application.

OCBC Bank — Automation Intern | 1 Year

- Developed Python automation solutions to improve operational efficiency, system monitoring, and reporting workflows.
- Built Selenium-based online banking monitoring automation to verify service availability before 5AM and generate email health reports with node-level diagnostics.
- Created file storage maintenance automation using Python and RegEx to delete targeted files and log files older than 90 days, reducing manual processing time by approximately 90%.
- Worked on CIF query performance testing, SSL certificate automation, database export tools, and database-to-Confluence automation.

National Service — HR Systems & Automation Development | 2024–2026

- Worked with SAP-based HR systems supporting personnel administration and operational tracking.
- Built Excel VBA HR dashboards integrating nominal roll and personnel data to filter eligibility based on PES status, IPPT validity, and combat shoot requirements.
- Developed Google Calendar auto-scheduling using databases and Google Apps Script, plus Telegram parade state bot for automated attendance reporting.
- Created internal chatbot to answer unit-related questions and reduce repeated manual queries.

Featured Projects

PulseWatch Monitoring Platform

- FastAPI-based backend monitoring platform that tracks uptime, response time, service status, and historical monitoring data.
- Implemented background monitoring workers using APScheduler, SQLite persistence, and Chart.js dashboard for response-time history visualization.
- Tech: Python, FastAPI, SQLite, APScheduler, Chart.js, HTML/CSS/JavaScript.

Food Donation Platform

- Full-stack platform connecting food donors with recipients through donation posting, request submission, and automated matching logic.
- Added machine learning demand prediction and Leaflet map visualization to display donation and request locations.
- Tech: Python, FastAPI, SQLite, scikit-learn, Leaflet, JavaScript.

Trading Bot with Telegram Alerts

- Python-based trading automation system integrated with Interactive Brokers API and Telegram notifications for trade activity and monitoring.
- Flow: Market Data -> Strategy Engine -> Trade Execution -> Telegram Alerts.

Object Recognition Tool

- Computer vision project detecting and classifying objects using image processing and machine learning techniques.
- Tech: Python, OpenCV, machine learning.

Selected Polytechnic Projects

- Verified Aid Resource Website: Grade A project helping users find verified aid resources such as laptops and subsidies; included forums and feedback functionality.
- CRM Fashion Website: multi-role system for Sales, Customer, Marketing Manager, and Product Manager views.
- UiPath Financial Workflow: automated extraction of P/E and stock metrics to support business report generation.
- Machine Learning: HR promotion prediction using KPI/training metrics and Airbnb price prediction using location data.
- Analytics: K-means clustering, hierarchical clustering, text analysis, and association rule mining.
- Other projects: Building Placement Game, Movie Ticketing System, Anime Gacha Website, Cruise SQL, Credit Card SQL, Flight SQL, MongoDB database projects.

Freelance & Side Projects

- Library overdue email automation system sending 600+ emails and reducing approximately 20 hours of manual work.
- Teacher workload matrix multiplier for workload allocation across subjects and academic levels.
- Telegram-based inventory ordering system with Google Sheets backend.
- Bicycle GMT schedule auto-updater, auto car depreciation calculator, Capex/Opex calculator, and law case time-balancing payroll system.

Certifications

- IBM certifications
- Scrum certifications
- UiPath certifications
- Workato certifications
- NLP, Machine Learning, Data Fluency, Artificial Intelligence, and Neural Networks certifications

Leadership, Awards & Activities

- Team lead across Polytechnic projects, OCBC internship projects, and National Service technology initiatives.
- Outstanding CFC recognition for leading technology development initiatives.
- Received director recognition letters and appreciation emails for automation contributions.
- Student leader; Secondary School Badminton Team; Polytechnic event support committee.

Links

GitHub: <https://github.com/JulienSLuk>

Portfolio: <https://juliensluk.github.io>

LinkedIn: <https://www.linkedin.com/in/julien-setiawan-lukman-532034211>